

Information-Geometric Correspondence (IGC)

A Field Guide to Excess Fisher Curvature Across Domains

1. Core Principle

Principle. Whenever a system is decomposed into subsystems, the deviation from local Fisher additivity defines an intrinsic curvature that encodes correlations, entanglement, or constraints.

Let g denote the Fisher information metric on a statistical manifold. For a partition $A|B$,

$$\mathcal{F} := g^{AB} - (g^A \oplus g^B).$$

- $\mathcal{F} = 0 \Rightarrow$ independent parts, flat geometry. - $\mathcal{F} > 0 \Rightarrow$ non-factorization, excess information, curvature.

The scalar contraction $\mathcal{R}_F := \text{tr}(g^{-1}\mathcal{F})$ is the *Fisher curvature*.

2. Minimal Equations

2.1 Additivity baseline

$$\mathcal{I}_{A \cup B} = \mathcal{I}_A + \mathcal{I}_B.$$

2.2 Excess Fisher curvature

$$\mathcal{I}_{A \cup B} = \mathcal{I}_A + \mathcal{I}_B + \text{tr}(g^{-1}\mathcal{F}).$$

2.3 Classical decomposition

For score functions S_A, S_B and residual R :

$$\mathcal{I}_{AB} = \mathcal{I}_A + \mathcal{I}_B + 2 \text{Cov}(S_A, S_B) + \text{Var}(R).$$

2.4 Quantum scaling

Separable: $I(\theta) \sim N$,
Entangled: $I(\theta) \sim N^2$.

2.5 Holographic duality

Boundary Fisher metric = Bulk canonical energy.

3. Operational Heuristics

- (a) Estimate KL or Fisher across partitions: non-additivity $\Rightarrow$ excess curvature.
- (b) If boundary relative entropy is preserved: interior measure-conjugacy is possible.
- (c) Divergence of $\mathcal{R}_F$: signals criticality or phase transitions.

4. Cross-Domain Examples

4.1 Physics

- Entangled probes yield excess QFI beyond separable limits. - Black-hole entropy area law: entropy $\sim$ surface, not volume. Boundary Fisher curvature encodes bulk geometry.

4.2 Neuroscience

- Neural codes exhibit synergy: joint Fisher info exceeds sum of parts. - Integrated information measures can be reframed as excess Fisher curvature.

4.3 Economics

- Market co-movements correspond to spikes in Fisher curvature. - Stress periods reveal super-additivity in covariance structure.

4.4 Climate/Earth systems

- Teleconnections (e.g. ENSO) manifest as $\mathcal{R}_F > 0$ across spatial regions. - Critical transitions marked by curvature divergence.

4.5 Machine Learning

- Feature interactions: positive $\mathcal{F}$ indicates synergy not captured by independent predictors.

5. Programmatic Steps

- Step 1. **Calculus of cuts:** develop transformation rules for $\mathcal{F}$ under coarse-graining and embeddings.
- Step 2. **Curvature budgets:** partition observed $\mathcal{F}$ into noise vs hidden geometry vs hidden drivers.
- Step 3. **Design by curvature:** maximize useful $\mathcal{F}$ (metrology) or minimize harmful $\mathcal{F}$ (stability).
- Step 4. **Correspondence library:** map domain-specific invariants (Bell, JLMS, Ruppeiner) to contractions of $\mathcal{F}$.

6. Outlook

The IGC provides a unifying framework: just as calculus gave a universal language for infinitesimal change, IGC proposes a universal language for *non-additivity*. Fisher curvature is its central invariant, linking statistics, quantum mechanics, and spacetime geometry.